

Mostafaa Abdelaziz

Halifax, Nova Scotia, Canada • 902-237-2694 • MostafaaAbdelaziz@dal.ca • LinkedIn • GitHub • Website

SUMMARY

Full-stack and mobile software developer with experience building web, mobile, and backend applications using React, React Native, Spring Boot, and Firebase. Strong background in authentication, REST APIs, real-time systems, and Agile development, with hands-on experience delivering production-style academic and client projects.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, TypeScript, Kotlin, SQL, HTML/CSS

Frameworks & Technologies: Node.js, React.js, React Native, Bootstrap, Spring Boot, OpenAI API, REST APIs, Expo, Docker

Databases/Cloud: MySQL, PostgreSQL, Firebase Firestore, Firebase Authentication, MongoDB, NoSQL, SQLite

Developer Tools: Git, GitLab, Postman, IntelliJ IDE, VS Code, Microsoft 365, JIRA, Wireshark, Linux/Unix

Coursework: Data Structures, Algorithms, Object-Oriented Programming, System Design, Mobile Development, Advanced Web Development, UX Design, Advanced Network Security

EDUCATION

Dalhousie University

Master of Applied Computer Science

September 2025 – Present

- CGPA: 4.19/4.3
- In progress: Certificate in Web and Mobile Computing

Bachelor of Computer Science with Distinction

September 2020 - May 2024

- CGPA: 3.83/4.3 (Dean's List and Sexton Scholar)
- Certification in Communication Technologies and Cyber Security

EXPERIENCE

Software Developer

January 2026 – Present

MOCEA LTD. (Client Project) | Halifax, Nova Scotia, Canada

- Built a React Native construction management application in a 4-developer team to improve issue tracking, inspections, scheduling, and on-site communication.
- Developed real-time multi-user workflows using Firebase Firestore synchronization for issue updates, reporting, and field coordination.
- Implemented secure authentication, push notifications, image uploads, and live data updates for mobile field operations.
- Integrated weather and maps APIs to support site risk awareness, task coordination, and location-based project visibility.
- Worked in an Agile environment with sprint planning, testing, debugging, and iterative feature delivery.

Teaching Assistant

September 2023 – January 2024

Faculty of Computer Science | Dalhousie University | Halifax, Nova Scotia, Canada

- Mentored students in network computing concepts, providing personalized guidance to improve their understanding and practical application of networking principles in assignments.
- Applied networking knowledge to provide clear and constructive feedback to students, actively supporting their learning and growth, and maintaining open, positive communication throughout the course.
- Conducted thorough assessments of Java code submissions, upholding course standards with an emphasis on accuracy.

PROJECTS

UniCarpool — Full-Stack Ride Sharing Platform (GitHub)

September 2025 – January 2026

React (TypeScript), Spring Boot, SQL, Docker, Android APK, JWT, OTP Authentication, UNIX, JUnit, Git, CI/CD Pipeline

- Architected and developed a full-stack ride-sharing platform supporting both carpool and taxi-style booking workflows.
- Built secure backend services using Spring Boot with JWT authentication and email-based OTP verification.
- Designed ride lifecycle features including ride creation, matching, booking, cancellation, and trip history.
- Developed a responsive React frontend integrated with REST APIs for real-time platform operations.
- Containerized and deployed the application on university infrastructure with Android APK support.

AI-Powered Learning Application (GitHub)

September 2023 – December 2023

Java, Kotlin, REST API, XML, Firebase, Git, Gitlab, JUnit, JSON, OAuth, OpenAI GPT-4, UNIX, CI/CD Pipeline

- Developed a seamless AI-assisted learning application for students and instructors as a tool to provide students the help they need to succeed in courses. The students can then track their grades which is automatically done.
- Created a business plan for which the application can also be used with the Free version (GPT-3.5) and a Premium version which uses (GPT-4 Turbo).
- Implemented a secure Login, Forgot Password and Register features that use OAuth security to maintain user's information using the Firebase Database which encrypts and hashes the users' credentials.
- Application provides students the ability to receive feedback from the instructor for quizzes, assignments and announcements done by the course instructor.